

No books, notes, computational devices, etc. are allowed. Use clear handwriting and justify your answers carefully. Please write your anonymous identification number on each sheet of paper.

1. Suppose R is a principal ideal domain. Let M be an R -module and let N be a submodule of M . The *isolator* of N in M is

$$\text{Iso}_M(N) = \{x \in M \mid \exists r \in R \setminus \{0\} \text{ such that } rx \in N\}.$$

Show that $\text{Iso}_M(N)$ is a submodule of M , and show further that every non-zero element in $M/\text{Iso}_M(N)$ is torsion-free. Finally, find an example of a proper non-trivial submodule that equals its isolator; i.e. $N = \text{Iso}_M(N)$.

2. Find the rational canonical form of the following matrix over $\mathbb{Q}$:

$$\begin{pmatrix} 2 & 4 & 0 \\ 1 & 4 & 1 \\ 3 & 0 & 3 \end{pmatrix}$$

3. Show that there are $\frac{1}{2}(k^n + k^{\lfloor (n+1)/2 \rfloor})$ neckties having n vertical stripes (of equal width), where each stripe can have one of k distinct colours. [The colour of each stripe can be seen on both sides of the tie.]
4. Let G be a finite group and $H \trianglelefteq G$ a normal subgroup. Let P be a Sylow p -subgroup of G , for some prime p . Show that $H \cap P$ is either $\{e\}$ or a Sylow p -subgroup of H (or both). Prove also that HP/H is a Sylow p -subgroup of G/H .
5. (a) Let $R = M_n(D)$ be the matrix ring over a division ring D , and for $1 \leq i, k \leq n$, denote by E_{ik} the matrix with 1 at the (i, k) entry and 0 elsewhere. Show that $M = RE_{kk}$ is a simple R -module.
(b) Give an example of a ring R and an R -module M such that M is not completely reducible.
6. Let R be a ring with unity and with the property that every non-empty set of left ideals of R has a maximal element. If R has no non-zero nilpotent ideals, show that R has no non-zero nil ideals.